V-OS

A graphical operating system for the Amstrad CPC 6128

Manual

Made from docs/manual/en/*.md and sdk/README.en.md by tools/mkdocs.py (make docs).

Contents

V-OS is a graphical operating system for the Amstrad CPC 6128, written from scratch in Z80 assembly. It has:

- a menu bar, draggable windows, dialogs and buttons, used with the keyboard or an AMX mouse
- three views for files and programs (Desktop, Launcher, Commander) and the settings (Setup)
- six languages as plugins: English, Greek, French, Spanish, German and Italian
- programs that run side by side and resume where they left off, a desk accessory and programs in the background
- Studio, a Scratch-style block editor that runs on the CPC itself
- an API with 42 calls and an SDK for programs in assembly

This manual is also in Greek (../00-index.md), French (../fr/00-index.md), Spanish (../es/00-index.md), German (../de/00-index.md) and Italian (../it/00-index.md).

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1. Getting started

What you need

- An Amstrad CPC 6128 or an emulator of it. V-OS is tested on Caprice32.
- The V-OS disc in drive A:, that is, the `build/vos.dsk` that `make` builds, and the programs' disc, `build/apps.dsk`, in B:.
- Optionally:
  - an AMX mouse
  - a second drive (B:)
  - a memory expansion (e.g. 512K), for more programs at once

Starting up

From BASIC:

```
RUN"VOS
```

V-OS loads the kernel, switches off the ROMs and takes over the whole machine. It detects the memory expansion by itself.

- **First view:** the Desktop. You can change it in Setup, on the START VIEW line.
- **Language:** English. If there is a `SETUP.CFG` on the disc, it starts with the language and settings saved there.
- **Exit:** FILE > EXIT TO BASIC (CONTROL+Q in the views). It asks for confirmation and restarts the CPC.

The screen

V-OS uses the whole screen (overscan): 384×272 pixels in mode 1 (4 colors) or 192×272 in mode 0 (16 colors).

- **Menu bar** (top): the menus of the view or program, usually FILE or the program's menu, VIEW and HELP. On the right is the keyboard language (EN, EΛ, ...).
- **Bottom line:** in each view, the keys that work there.

Keys that work everywhere

Key	What it does
F1	Desktop
F2	Launcher
F3	Commander
F4	Setup
F5	opens and closes the desk accessory (ACCESSORY)
F6	the running programs, to choose one
ESC	opens the first menu. In the menus, ←/→ change menu, ↑/↓ choose an item, RETURN runs it, ESC closes
CONTROL+letter	the menu shortcuts (e.g. ^Q, ^L)

CONTROL+L	switch language (VIEW > SWITCH LANGUAGE)
CONTROL+K	the dialog with the keys (HELP > KEYS)
CONTROL+SHIFT	switches the keyboard: English, the pack's, and, with KEYBOARDS: ALL SIX in Setup, the other languages'
TAB, SHIFT+TAB	move the focus between buttons. RETURN, ENTER or SPACE press the button that has the focus

F1–F6 are the f1–f6 keys on the CPC's numeric keypad.

The pointer and the mouse

- **With an AMX mouse:** the pointer moves with the mouse, and its button clicks. The mouse is switched on and set up in Setup (AMX MOUSE, AMX READING).
- **Without a mouse:** the pointer moves with the arrow keys, and COPY clicks. In views that use the arrow keys for choosing (Desktop, Launcher, Commander), the arrows choose files and programs instead of moving the pointer.

Most things work with both: one click on a file or program selects it, and a second click opens it.

Windows and dialogs

- **Dragging:** every window can be dragged by its title bar: the dialogs, the Desktop's FILES window, the desk accessory, and the windows of block programs. While you hold the button a dashed outline is shown, and the window is redrawn where you let go.
- **Dialogs:** wait for an answer. TAB and ←/→ change the button, RETURN presses it, and ESC chooses the last one (usually NO or CANCEL).
- **Text input** (e.g. renaming): you type at the end, DEL deletes, RETURN accepts and ESC cancels. You can also type in the language of the pack.

Discs and files

- **Format:** V-OS reads and writes AMSDOS discs (DATA or SYSTEM format) in A: and B:. BASIC can read its files too.
- **Names:** 8 characters plus 3 for the type (`NAME.TYP`).

Types you will come across:

Type	What it is
`.BIN`	a V-OS program (or a binary file)
`.VBN`	a V-OS program made with blocks
`.VBS`	a Studio program (its source)
`.OVL`, `.MSG`, `.LNG`	parts of V-OS itself: views, texts, languages
`.SYS`	system files (e.g. `VBRUN.SYS`, the Studio player)
`SETUP.CFG`	the settings

The settings, languages and views are read from A:, so the V-OS disc must stay there.

Error messages

Disc errors are shown in a dialog: DRIVE NOT READY, WRITE PROTECTED, FILE NOT FOUND, DISC FULL, DIRECTORY FULL, NAME ALREADY USED, INVALID NAME and others.

Two more concern programs:

- **NOT A V-OS PROGRAM:** the file does not have the header of a program.
- **CLOSE ANOTHER PROGRAM FIRST:** there is no room. See chapter 3.

2. The views

V-OS has four "views": ways to see the files and programs, and the settings. You switch view with F1–F4 or from the VIEW menu. Each view remembers which file was selected, and the next one selects the same file.

The common menus

The last two menus are the same in all views and in programs:

VIEW		HELP	
DESKTOP	F1	KEYS	^K
LAUNCHER	F2	ABOUT V-OS	
COMMANDER	F3		
SETUP	F4		
PROGRAMS (running)	F6		
SWITCH LANGUAGE	^L		

HELP > ABOUT V-OS shows the version (0.15 beta; each milestone adds 0.01), the date of the build, the maker (Revive8bit, 2026) and the architect (Vasper).

Desktop, F1

Mode 1. A FILES window shows the files of a drive as icons, four in each row. The type is written inside the icon and the name below it, and a `.VBN` has a "play" mark next to it. On the right, two icons choose A: or B:. The title bar shows the disc format and the number of files, and the free space is shown at the bottom.

Key	What it does
arrows	choose a file (↑/↓ one row)
TAB	the next drive: A:, B:, DISK (with an M4 board)
RETURN, ENTER	open: a program runs, a file of a known type opens in its program, others show their info
^C	copy to the next drive (from DISK: to A:)
^D	delete (asks first)
^N	rename
^R	read the discs again
mouse	click on a file: select, again: open. Click on a drive: opens it

- **FILE menu:** OPEN, FILE INFO (^I), COPY TO NEXT DRIVE, DELETE, RENAME, READ DISCS AGAIN, KEEP THE DESKTOP (^S), EXIT TO BASIC.
- **KEEP THE DESKTOP** (^S) writes the files window as it is (where, its size, its drive, or closed) into `A:SETUP.CFG`: at the next start the Desktop opens it so at once. Setup's ^S keeps it too.
- **The FILES window** can be dragged by its title and stays there, even when you switch view.
- **At first there is no window and no drive is chosen:** the desk (and its picture) shows. A drive (a click on its icon, or TAB) opens its window; the X on its title closes it, and so does a click on the open drive. The small square in the corner below on the right makes the window larger or smaller in whole icons (2–5 in a row, 1–5 rows; four by five at first), and the bar on the right shows which part of the list is in view: a click above or below its dark part scrolls a window's worth up or down. The size is kept, like the place.

- **Opening by type** (also in the Commander): `.TXT` files open in NOTEPAD, `.MD` in MARKDOWN, `.PIC` and `.LYR` in PAINT, `.VDB` in DATABASE, `.VSH` in SHEET, `.MUS` in MUSIC, which starts playing them. The program is looked for on the file's drive, then on A:, then on B:. If it is not found, "NO PROGRAM FOR THIS FILE" is shown. If the program has unsaved changes, it asks first.

Launcher, F2

Mode 0, 16 colors. Shows only the programs of one drive (`.BIN`, `.VBN`, `.BAS`), A: at first, as tiles in a 3×3 grid, with the next pages below. The page line starts with the drive (`A:`, `B:`). Every program that has an icon shows it on its tile. The selected tile is white.

Key	What it does
arrows	choose (↑/↓ one row. Past the edge, the next or previous page)
RETURN, ENTER	runs the program. If it is already running, it resumes where it was
^A	loads it as the desk accessory (AS ACCESSORY). F5 then opens it
^R	read the disc again
TAB	the next drive (NEXT DRIVE): A:, B:, DISK. A program started from B: comes back to B:
mouse	click: select, again: run

FILE menu: RUN, AS ACCESSORY, NEXT DRIVE, READ DISCS AGAIN, EXIT TO BASIC.

Commander, F3

Mode 1, two panels with file lists: A: on the left, B: on the right. Each panel shows the format and the number of files in its title, and the free space at the bottom. The active panel has a dark title and a selection.

Key	What it does
↑/↓	choose (with SHIFT: one page)
TAB, ←/→	the other panel
RETURN, ENTER	open, as in the Desktop. The info (size, type, load address, the first bytes) with ^I
^C	copy to the other panel, of any size. If there is a file with the same name, it asks first
^D	delete
^N	rename
^R	read both discs again
^G	the next drive for the active panel: A:, B:, DISK (also a click on its title)
^F	find files: a name with `*` and `?` (`*.TXT`, `README.*`, `NOTE?AD.BIN`; without a dot, any extension), looked for on A:, B: and DISK (the folder it is in). The active panel lists what was found, up to 64 files, each with its drive (`B:README.TXT`). RETURN goes to the file on its drive; ^C, ^D, ^N and ^V work on the found files as they are; ^R or ^G ends the list
^V	show the text: the chosen file's first lines in the other panel, until it is drawn again. A picture (`.SCR`, as PAINT and SHEET's CHART save them) shows on the whole screen in mode 0 with its own inks; a key or a click brings the Commander back
^E	copy the disc: the whole disc of the active panel onto the other panel's (A: and B:), track by track, each track formatted first as the source's (DATA or SYSTEM). It asks first: the other disc's files are lost
mouse	click: select, again: open

FILE menu: OPEN, FILE INFO (^I), SHOW THE TEXT (^V), FIND FILES (^F), COPY THE DISC (^E), COPY TO OTHER PANEL, DELETE, RENAME, READ DISCS AGAIN, NEXT DRIVE (^G), FORMAT DISC (DATA),

EXIT TO BASIC.

> FORMAT erases all the files on the disc of the active panel. It asks first.

DISK: the M4 board's SD card

With an M4 board, the Desktop has a third drive icon, DISK, under A: and B:, and TAB goes A: → B: → DISK. The Launcher and the Commander's panels (^G) reach it too. The title shows the folder you are in, e.g. `DISK:/GAMES/`.

- **Folders** show as folders (an icon with a tab, or DIR in the Commander). RETURN on one goes into it, and `..`, the first entry, goes back up. In the Launcher, folders are there too, so that you can reach the programs inside them.
- **Everything as on a disc:** programs run from it, files open in their programs, and ^C, ^D and ^N copy, delete and rename. Copies between DISK and a disc go up to 16K. Folders and files are named as on the CPC (8.3: `NAME.EXT`); names that are longer do not show.
- **M4.SYS** (on `vos.dsk`) is loaded from A: the first time DISK is used.
- **The time and the dates:** the menu bar shows the M4's time ("09:25", left of the keyboard's language) in every view and program that has room for it, and FILE INFO (^I) gives a file's date and time on the card under its size. The time comes from the M4's time server (set with `NETSET` on the CPC).
- **In the programs:** OTHER DRIVE (^W in SHEET) goes A: → B: → DISK, and the bottom line says `DISK:NAME.TXT`. OPEN's list shows the folders too: RETURN on one (or on `..`) lists that folder, and SAVE writes in the folder you opened from. READER writes the card as `C:` in its places.
- **Databases** go on DISK too: DATABASE's ^W reaches it, and STREETS run from the card keeps its STREETS.VDB there.
- **Not yet:** the SD card is not formatted from V-OS. V-OS itself still starts from a disc in A: (type `|DISC` before `RUN"VOS"`: with an M4 board, the CPC starts on the SD card).

Setup, F4

Mode 1. One line for each setting. Changes take effect at once. When saved (RETURN or ^S) they are written to `A:SETUP.CFG` and also apply at the next start-up.

Setting	Values
START VIEW	DESKTOP, LAUNCHER, COMMANDER
LANGUAGE	ENGLISH, ΕΛΛΗΝΙΚΑ, FRANÇAIS, ESPAÑOL, DEUTSCH, ITALIANO
REPEAT DELAY	how long you hold a key before it starts repeating: SHORT, MEDIUM, LONG
REPEAT SPEED	SLOW, MEDIUM, FAST
AMX MOUSE	ON, OFF
AMX READING	LEVELS (at first) or PULSES: how the mouse is read. LEVELS counts every read with movement, PULSES only the starts of the pulses. If the pointer moves too fast or too slowly, try the other one
DESK COLOUR	BLUE, BLACK, GREEN, RED
KEYBOARDS	EN + PACK, ALL SIX (the ones CONTROL+SHIFT goes through)
THEME	LIGHT, DARK (bars, menus, dialogs, windows and lists in black with white text; in programs, their menus and dialogs)

BACKGROUND	NONE, or a `.SCR` picture: LEFT/RIGHT go through NONE, A:, B: and DISK, and on a drive its `.SCR` files are listed to choose one. The Desktop shows it in the middle of the desk, the desk colour around it; a PAINT picture (mode 0, 16 colours) in the desk's four colours. It needs a memory expansion (the last 16K bank of it keeps the picture); without one: NEEDS A MEMORY EXPANSION. After three minutes with nothing done on the Desktop, the picture shows alone (a screen saver); any key or the mouse brings the Desktop back

Key	What it does
↑/↓	choose a setting
←/→	change the value (also a click on the line)
RETURN, ^S	save

Changing the LANGUAGE loads the language pack from A: at once. See chapter 4.

3. Programs

Starting and ending

A program starts in one of these ways:

- from the Launcher: select it and press RETURN, or click twice
- from the Desktop or the Commander: FILE > RUN
- from another program, such as Studio, which runs whatever you make

Each program has its own menu next to the common VIEW and HELP. It ends from its menu (usually BACK or STOP, CONTROL+Q) and returns to the view it was started from.

Several programs at once

V-OS programs are of two kinds:

- **Small, with a relocation table:** DEMO, MIXED and CLOCK, the web editor examples (PONG, CATCH, STARCAT, COUNTER), and whatever the SDK makes. They share 12K of memory, and as many as fit run together.
- **Programs that need all the memory:** STUDIO and whatever Studio makes (CALC, MINES, FLAP, TICKER and your own). So are MARKDOWN and READER, which are over 12K. Without a memory expansion they run on their own. If one is already running, you get "CLOSE ANOTHER PROGRAM FIRST".

When you go to a view (F1–F4) while a program is running:

- A small program stays in memory. You pick it up again from the Launcher (RETURN on it) or with F6, and it resumes exactly where it was.
- A program that has all the memory ends, unless there is a memory expansion.

F6 (VIEW > PROGRAMS) shows the list of running programs. You choose one with the arrows and RETURN. If none is running, you get "NO PROGRAM IS RUNNING".

With a memory expansion (e.g. 512K, which V-OS finds by itself at start-up), every program that needs all the memory gets its own bank. So STUDIO, CALC, DEMO and PONG can all run together. Since M43 every program also gets a data bank of its own, 16K, when it needs one: what it keeps there is not lost while other programs or views run.

In the background

A program behind another one, or behind a view, normally **freezes** and resumes when it comes back to the front.

Some programs ask to **keep running** in the background too, e.g. TICKER, which counts time. While there they do not draw or play sound. When they come back to the front, they have also counted the time they were away.

In Studio you choose this for your program with STUDIO > IN BACKGROUND.

MUSIC is the exception: its song plays from the interrupt, 50 times a second, so it goes on evenly behind everything. It stops only for a moment while the disc is read.

Windows

With a **memory expansion**, CLOCK, SYSINFO, DIARY, TELNET, MAIL, FTP, WEBGET and DATABASE open in a **window** on the desk, over the windows of the other programs that are running. So does a Studio program in mode 1 that opens a window of its own (such as CALC): the window shows its inside, with its buttons. DIARY, TELNET, MAIL, FTP, WEBGET and DATABASE show their screen as it was in the window; made smaller, it shows the top left part of it. NOTEPAD, MARKDOWN, READER and SHEET open in a window too and lay their text out in it: as many columns and rows as fit, the status line under them, and again whenever you resize it. Their page (^M), and SHEET's chart, show full screen; ^M again brings the window back where it was, as large as it was. Each window keeps a copy of itself in memory, so when you move a window over the others they come back at once, without their programs drawing them again. The Desktop's picture (BACKGROUND in Setup) shows under the windows. The Desktop's files window and drive icons are among the windows too: they show behind the programs, and on the Desktop (F1) the programs' windows show on its desk. A click on a window brings it to the front, and a click on the Desktop's window or icons brings the Desktop. A window's X asks its program to close as ^Q does, so NOTEPAD, for one, asks first about unsaved changes. A program in a window does not change the screen's colours. TELNET, MAIL and FTP in a window go on receiving while behind, if no other window covers them. The accessory (F5) and the list of programs (F6) open over the windows. The other programs still open full screen.

- The menu bar stays at the top and is the bar of the window **in front**. A click on another window brings it to the front, with its bar.
- Drag a window's **title** to move it, and its **bottom right corner** to change its size. The program lays out what it shows for the new size.
- The **+** in the title, or **F7**, gives the program the whole screen (under the bar), and back to a window. **X** closes the program.
- The clock **keeps ticking** while another window is in front, as long as none covers it.

The desk accessory (ACCESSORY)

A small program, such as CLOCK, can open in a window **on top of anything**: a view or a program.

1. In the Launcher, select CLOCK and choose FILE > AS ACCESSORY (^A).
2. From then on, **F5** opens and closes it, wherever you are. ESC also closes it.

While it is open, it gets the keys and clicks. Its window can be dragged by its title. When it closes, whatever was underneath is redrawn.

The page (mode 2)

NOTEPAD, MARKDOWN, READER and SHEET have PAGE (MODE 2) in their menu, ^M. The screen goes to mode 2: black text on white paper, 768 pixels across, with the menus and dialogs black on white too. The editors (NOTEPAD and MARKDOWN) show 80 columns between the page's edges, a line where each page of 60 lines ends, and PAGE n in the last line. MARKDOWN's preview and READER show the page in 80 columns with its styles. SHEET shows 10 columns (A–J) with lines between them. ^M again goes back to mode 1.

The programs on the disc

A CPC disc holds 180K, so V-OS comes on two. `vos.dsk` (A:) has the system, DEMO, MIXED, STUDIO and CLOCK, and the `.VBS` sources of the VB programs, which Studio opens from A:. `apps.dsk` (B:) has the others, from PONG to WEBGET: in the Launcher, TAB shows B:. Some programs have a disc of their own, as the table says.

Program	What it is

STUDIO	the block editor: you make programs on the CPC (chapter 5)
DEMO	the first V-OS program: a square that moves by itself. The arrows steer it and SPACE changes its color
MIXED	the "mixed screen": a band with 16 colors (mode 0) between mode 1 bands. A click shows the coordinates
CLOCK	the time since V-OS started; with an M4 board the real time and, as a program, the date (from the M4's time server, set with `NETSET` on the CPC). It also runs as a desk accessory
CALC	a calculator in a window with buttons, made in Studio (there is also `CALC.VBS`)
MINES	an 8×6 minesweeper with 8 mines, made in Studio. A click opens a cell, and NEW starts a new game
BRICKS	brick breaking: a paddle with ←/→, 32 bricks, 3 lives, levels with an ever faster ball. At GAME OVER, SPACE for a new game
VROUVES	tower defence, from the CPC game LET'S GO FOR VROUVES: 7 stages, from the Kafenio to the cemetery, with 10 waves each. It is on its own disc, `vrouves.dsk`: put it in B: and press TAB in the Launcher. Arrows move the cursor, 1–6 picks a defender, SPACE builds, S sells, N calls the next wave, P pauses. Points for every invader and for the villagers left at the end of a stage. The disc keeps `VROUVES.SAV`: CONTINUE starts from the stage you reached, HIGH SCORES shows the best five with their names. Written with the SDK (`myapps/vrouves`)
NOTEPAD	a text editor in mode 1, 48×22, for files up to 4K, or up to 16K with a memory expansion: then it keeps about 2.5K around the cursor in its memory and the rest in its own data bank, and SELECT ALL takes that part. NEW, OPEN (only `.TXT` files), SAVE, SAVE AS, OTHER DRIVE. Arrows, DEL, CLR and RETURN; a click places the cursor. The bottom line shows the line and the column. EDIT menu: COPY ^C, CUT ^X, PASTE ^V, SELECT ALL ^A; SHIFT and the arrows select. The clipboard (up to 1K) is shared by all programs. PRINT ^P prints the text on a printer. UNDO ^Z (EDIT menu) takes back the last change: what was typed or pasted, or what DEL, CLR, CUT or typing over a selection took out (up to 256 characters); ^Z again does it again. FIND ^F (EDIT menu) looks for a text from the cursor on, in all of the text (A–Z without case), and selects it; FIND AGAIN ^G the next one; REPLACE ^R asks for the text and what to put instead, and changes it in all of the text (NOT FOUND if it is not there) (`myapps/notepad`)
PAINT	drawing in mode 0 with 16 colors: a picture of the whole CPC screen, 160×200 without the overscan (`.SCR`, 16K, which BASIC and other CPC programs load too; without a memory expansion 128×128, `.PIC`), or 64×64 at 2× zoom in 3 layers (`.LYR`, NEW LAYERS). Tools: P pencil, B brush, L line, X box, F fill, C circle (from its centre to where the button comes up); TAB changes the color. Arrows and COPY draw without a mouse. The window shows 128×128 of the picture: the bars on its right and under it show which part, and a click or a drag on them moves it. Under the colors: where the pointer is in the picture, and VIEW, the window's top left. Layers: 1–3 choose one, H hides or shows it, U and D move it up and down, E is the eraser. UNDO ^Z takes back the last tool, and ^Z again brings it back (FILL on the whole screen cannot be taken back; after another program has run there is nothing to take back). EXPORT .SCR (for the 128×128 picture and the layers; the whole screen is a `.SCR` already): the picture (the layers as they show) as a CPC screen, a 16K `NAME.SCR` (mode 0, in the middle, with the palette), for BASIC or other programs. In BASIC: `MODE 0:LOAD"NAME.SCR",&C000:FOR I=0 TO 15:INK I,PEEK(&C7D0+I):NEXT:BORDER PEEK(&C7E0)` (`myapps/paint`)
MUSIC	plays Arkos Tracker 3 songs (`.MUS`, made by `make songs`), with three volume bars. It is on its own disc, `music.dsk`, in B:. OPEN, SPACE plays and stops. The music goes on behind other views and programs, evenly, even while another program loads or draws: it plays from the interrupt. Studio's MUSIC block starts it behind a program with a song (`myapps/music`)

MARKDOWN	an editor for Markdown files (`.MD`, up to 6K), like NOTEPAD. ^P shows the page: `#` headings, **bold**, *italic*, `code`, lists, quotes, lines, code blocks and underlined links, without the marks. Lines that follow each other make one paragraph (two spaces at the end break the line). The arrows scroll the page, RETURN or ^P go back to the text. It is on its own disc, `markdown.dsk`, with a `README.MD`. EDIT menu: COPY ^C, CUT ^X, PASTE ^V, SELECT ALL ^A; SHIFT and the arrows select. The clipboard (up to 1K) is shared by all programs (`myapps/markdown`)
DIARY	days and what is on them: the month in mode 1, a week a row from Monday, the chosen day inverted and the days with notes marked `*`, and under it the chosen day's notes (up to 10, of 40 characters). The arrows move by a day and a week, `<` and `>` by a month, TAB goes to the notes (UP/DOWN choose one, RETURN changes it, DEL deletes it). ^A adds a note, ^T goes to today. With an M4 board it starts on today; without one on the last day with a note. The notes are kept in `DIARY.VDB` on the drive it was started from (^W the other drive), made the first time, through `DB.SYS` (`myapps/diary`)
SHOW	the `.SCR` pictures of a drive one after the other, in mode 0, with their own colours (PAINT's): the next one every five seconds, → and ← by hand, SPACE stops and goes on, ^W the other drive (A:, B:, DISK). Up to 48 pictures (`myapps/show`)
SYSINFO	shows the V-OS and API versions, the memory (all of it and what is free for programs) and the free space on the discs in A: and B: (^R again). An example of `VOS_EXT` (`myapps/sysinfo`)
TELNET	text over the network with the M4 board: ^O CONNECT asks for `HOST:PORT` (a name such as `bbs.example.org`, or numbers such as `192.168.1.10`; without a port 23) and connects by TCP. What comes is written in the window, which scrolls up at its end; colours and other ANSI codes are left out, Greek in ISO 8859-7 shows as it is, and the telnet options the other end asks for are refused. The line typed at the bottom goes with RETURN. ^D CLOSE ends the connection, as the other end may; the first line shows the state (CONNECTED, NO CONNECTION...). Without an M4 board: NO M4 BOARD (`myapps/telnet`)
MAIL	e-mail with the M4 board: ACCOUNT asks for the POP3 and SMTP servers (`HOST:PORT`; without a port 110 and 25), the user, the password and your address, and keeps them in `MAIL.CFG` beside MAIL. ^G GET MAIL lists the newest 20 messages, the newest first (who from, the subject); RETURN reads the one chosen (UP/DOWN a line, LEFT/RIGHT a page, RETURN or DEL the list): quoted-printable and base64 are decoded, UTF-8 and encoded subjects become Greek, and of a message in parts its text is shown. ^S SAVE AS TEXT writes it as a `.TXT`, ^D DELETE deletes it on the server (when the connection ends). ^W WRITE asks for TO and SUBJECT and sends a `.TXT` of the drive shown (written in NOTEPAD; OTHER DRIVE changes it), and ^R REPLY answers the one you read with "Re: ". Plain connections only (no SSL/TLS), so a server of your own or one that allows them; the password is kept as it is (`myapps/mail`)
FTP	files to and from an FTP server with the M4 board: ^O CONNECT asks for `HOST:PORT` (without a port 21), the user (none: anonymous) and the password, and lists the server's folder: folders as `<DIR>`, files with their sizes. UP/DOWN choose, RETURN opens a folder or gets a file, DEL (or ^U) goes up a folder. ^G GET THE FILE saves it (up to 16K, under an 8.3 name it asks for) on the drive shown (OTHER DRIVE), ^P PUT A FILE sends one of that drive, and ^D DELETE THE FILE deletes it on the server. ^K CLOSE ends the connection; the first line shows the state and the folder (`myapps/ftp`)
WEBGET	downloads files from the web with the M4 board: GET A FILE (^G) asks for an address (HTTP, e.g. `example.com/index.html`) and a name, and the file (up to 16K) is saved on the drive it shows (OTHER DRIVE). It also shows the network status. Without an M4 board: "NO M4 BOARD" (`myapps/webget`)
READER	reads Markdown pages (`.MD`, up to 6K) from the disc or from the web with the M4 board, shown as in MARKDOWN's preview. TAB selects the next link (inverted), RETURN opens it, DEL goes back to the previous page (it keeps 4). A link without `http://` opens next to the page, on the disc or on the same site. OPEN A FILE and OPEN AN ADDRESS (^U, HTTP, e.g. `example.com/README.md`). `https://` addresses cannot be opened. It is on its own disc, `reader.dsk`, with `INDEX.MD` and `ABOUT.MD` (`myapps/reader`)

STREETS	a street index in a V-OS database: type the first letters of a street and see the streets that start so, with their area and post code; DEL takes a letter back. ^A adds a street, ^D deletes the first one in the list. The streets are in `STREETS.VDB` on its disc, `streets.dsk`, in B:; the first time it makes it, with streets of Athens. An example of the database API (`myapps/streets`) Finding ignores capitals, small letters and accents.
DATABASE	databases, like a small Access: ^N a new database (`NAME.VDB`), ^O open one, ^T a new table (its name, and fields: a name, TEXT up to 40 characters or NUMBER 0–65535, and whether it is sorted by the first one). The list shows a record a row and the fields in columns; the arrows choose and scroll, TAB goes to the next table, and if the table has a key, what you type goes to the records that start so (without case or accents). RETURN opens the form: a field a row, RETURN changes it, ^S saves, ^B back to the list. ^A a new record, ^D delete, ^W the other drive, ^P prints the table as the list shows it, every record. PRINT LETTERS ^L prints a `.TXT` of the drive (a letter written in NOTEPAD) once for each record of the table, a page each: a field's name in braces, `{NAME}`, becomes what the record has in it (without case; an unknown name stays as it is). IMPORT CSV adds a record for each line of a `.CSV` to the table in use: the fields in order, by commas or semicolons, texts cut to their length, numbers from their digits; a first line that starts with the first field's name (the heads) is left out. It opens STREETS's `.VDB` too (`myapps/database`)
SHEET	a spreadsheet: 16 columns (A–P) and 64 rows, 5 columns and 19 rows on the screen. A cell holds, up to 40 characters, a number (`12`, `-3.5`; the decimal point is a dot), text, or a formula after `=`: `+ - * /`, brackets, cells (`B3`), SUM, AVG, MIN, MAX and COUNT of cells and ranges (`=SUM(B2:B9)`), ABS, INT, ROUND(x, 0–2). Numbers have two decimals, up to ±21474836.46. The arrows move the cursor, RETURN changes the cell, typing starts a new text, DEL (or CLR) empties it; after each change all the formulas are worked out again. The first line shows the cell's text. A cell with an error shows `#ERR`, a number that does not fit `########`. ^N new, ^O open, ^S save, ^A save as (`NAME.VSH`, a text file, a line a cell: `B3 =B1*2`), ^W the other drive, ^P prints the sheet as the page shows it, columns A–H and then I–P, up to the last row used. ^G CHART: a range of a column or a row (`B2:B9`, up to 16 cells; at first the cursor's column down to its last number) as bars in mode 0, each as high as the cell is to the largest, with the texts of column A (or row 1) under them; ^S saves the chart as a `.SCR` (the bars, without the texts), which PAINT and BASIC open, any other key goes back. OPEN CSV reads a `.CSV` (from a PC spreadsheet, or a database): a line a row, the fields from column A, separated by commas or semicolons (the first line shows which); fields in quotes may hold them, and with semicolons `1,5` is the number 1.5. Saved, it becomes a `.VSH` of the same name. The formulas are worked out by `CALC.SYS` on A:, which any program can use (`VX_CALC`). It is on its own disc, `sheet.dsk`, with the example `BUDGET.VSH` (`myapps/sheet`)
PONG	a ball and a paddle with the arrows, score and lives
CATCH	catch the falling stars
STARCAT	a cat with costumes that walks, a star, a melody from a list
COUNTER	a counter with +1, −1 and 0 buttons, in mode 1
FLAP	a bird that flaps its wings (costumes), and a star that changes costume every second
TICKER	counts seconds and keeps going in the background

- Block programs are `.VBN` files. They have a PROGRAM > STOP menu (CONTROL+Q).
- Those that also have a `.VBS` (CALC, MINES, FLAP, TICKER) open in Studio so you can look at them or change them.
- `VBRUN.SYS` is not a program: it is the "player" that Studio needs.

Printing

NOTEPAD, SHEET and DATABASE print with ^P (PRINT in their menu) on a printer connected to the CPC's printer port (Centronics), such as the Amstrad DMP2000 or an Epson-compatible one.

- **Seven bits:** the CPC's port sends 7 bits, so letters over 127 go out as plain letters: accents are left out (É → E, ñ → n, ß → s) and Greek is written in Greeklish, a letter for a letter (Α → A, Θ → 8, Ξ → 3, Ψ → 4, Ω → W).
- **Pages:** each page starts with a heading, the file's name and the page's number (`NOTES.TXT PAGE 2`), and after 60 lines a new page begins by itself.
- **The end:** each printout ends with a new page.
- **No printer:** if the printer stays busy for two seconds (none connected, off line, no paper), THE PRINTER IS NOT READY appears and the printing stops.
- Programs print with the `VX_PRINT` call (chapter 7).

4. Languages

V-OS speaks six languages. English is built in. The other five are **language packs**, files on the disk that bring three things: all the system texts, the keyboard, and the letters (font) of the language.

Language	File	Keyboard
English	built in (`ENGLISH.MSG`)	the CPC's English one
Ελληνικά	`GREEK.LNG`	Greek, with accents
Français	`FRENCH.LNG`	AZERTY, digits with SHIFT, dead keys ^ and ¨
Español	`SPANISH.LNG`	Ñ on the `:` key, dead keys ´ ¨ ` ^, and ¡ ¿
Deutsch	`GERMAN.LNG`	QWERTZ, Ü Ö Ä ß, dead keys ´ and `
Italiano	`ITALIAN.LNG`	è é ò à ù ì on their own keys

Switching language

- **Permanently:** in Setup (F4), the LANGUAGE line cycles through all the languages. Each one loads from A: at once, and the whole screen is redrawn in it. Saving (^S) keeps it for the next startup too.
- **Quickly:** CONTROL+L (VIEW > SWITCH LANGUAGE) switches between English and the language of the pack that is loaded.
- **Keyboard only:** CONTROL+SHIFT switches the keyboard. The menu bar shows on the right which one is active (EN, EΛ, FR, ...). With Setup's KEYBOARDS at EN + PACK it goes between the English one and the pack's; at ALL SIX it goes through all of them: EN, the pack's, the other four, and EN again. So you can type French with the French keyboard while the system is in Greek. Every pack holds the keyboards of all of them.

The symbols `@ # [ ] { } \ |` are not on the packs' keyboards, because the CPC has no AltGr. Type them with the English keyboard.

Program texts

Programs have their own texts in **two languages**: English and, optionally, one more (today, Greek). A program appears in its second language only when the pack is in that language. Otherwise it appears in English. So, with French, Studio appears in English. Studio's programs, though, can have their texts in all six languages (chapter 5): CALC is called "CALCULATRICE" in French.

One alphabet for all

All languages write with the same character set, the **V-OS set**, and every pack has all its letters. So a text or a database can mix Greek, French, German, Spanish and Italian, and they show right whatever language is active: in Setup, LANGUAGE shows ΕΛΛΗΝΙΚΑ, FRANÇAIS, ESPAÑOL with their own letters.

Codes 128 to 255 are the Greek ones of ISO 8859-7, as they were, and the codes it leaves free hold À Á Â Ä Ç È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Ö Ù Ú Û Ü, their small letters, ß and ¡ ¿. The ´ is written as the Greek tonos (΄) and µ as µ. There is no ÿ, Œ or œ. The table is in `tools/voscs.py`. A Greek text file from a PC (ISO 8859-7) reads as it is; a French or German one in Latin-1 needs converting.

With the English keyboard, SHIFT and `^` gives £.

Files in **UTF-8**, from a PC or the web, are opened by NOTEPAD and READER, which turn them into

the V-OS set; what it has not (Chinese, say) becomes `?`. On the PC, `tools/voscv.py` converts files to and from V-OS (UTF-8, Latin-1 and others) and says what a file is (`check`).

Making a new language

The language sources are in `lang/<code>/`:

- `messages.txt`: the system texts. The base list is defined by `lang/en/messages.txt`, and anything missing from a pack appears in English.
- `keymap.txt`: the keys that differ from the English keyboard.
- `pack.txt`: the code, the name, the character set and the font.

`tools/mklng.py` turns them into a `.LNG`, and the Makefile puts it on the disk. All packs have the same font, `lang/vos/font.txt`, and `charset vos`.

5. Studio: programs with blocks

STUDIO is a Scratch-style block editor that runs on the CPC itself.

- You make sprites and scripts out of blocks.
- The program runs at once with CONTROL+R.
- It becomes a normal V-OS program (`.VBN`) that also starts from the Launcher.

The screen

Studio uses a "split" screen, with two zones:

- **Top, in mode 0 (16 colors):**
  - **left, the stage** at half size
  - **right, the SPRITES panel:** the sprites with their color and position, the size of the selected one (W, H), the background and the program's mode (e.g. `STAGE 2 M1`), and the file name
- **Bottom, in mode 1:**
  - **left, the palette** with the blocks of one category (`< EVENTS >`)
  - **right, the script**, one line per block
  - **at the end, the status line**, with the keys that work where you are

TAB goes from the script to the palette, then to the stage, and back to the script.

Menus

STUDIO		EDIT
NEW	^N	DELETE LINE
OPEN (`.VBS`)	^O	+ SPRITE (up to 6)
SAVE	^S	− SPRITE
SAVE AS		BACKGROUND (the stage's color)
RUN	^R	SCREEN MODE (0 or 1)
IN BACKGROUND		TEXT LANGUAGE (^T)
BACK	^Q	

- **RUN (^R):** saves `NAME.VBS`, writes `NAME.VBN` and runs it. The program stops from its own menu (PROGRAM > STOP, ^Q), and then Studio starts again with the same program open.
- **IN BACKGROUND:** asks whether the program keeps running when it is behind another one or behind a view, or pauses, which is the default.
- **SCREEN MODE:**
  - **Mode 0:** 16 colors, 192 pixels wide.
  - **Mode 1:** 4 colors (pens 0–3), 384 pixels wide, and more text.

The palette

←/→ change the category, ↑/↓ choose a block, and RETURN adds it below the script line. A click on a

palette block does the same.

A block that opens (FOREVER, REPEAT, IF) goes in together with its END, and is deleted together with it. A block with text asks for the text at once.

The script

Key	What it does
↑/↓	line
←/→	slot of the line (the slots are the values inside the block)
RETURN	with no slot selected: adds the palette block. On a text slot: change the text. On a key slot: waits for you to press the key
CLR	deletes the line
DEL	deletes the last digit of a number
mouse	click on a slot: selects it. Second click: changes its kind

The program is a list of lines:

- Each `WHEN …` starts a script, for a sprite or for the stage. The script runs up to the next `WHEN`.
- The indentation comes by itself from the blocks that open and close.
- Lines before the first `WHEN` are ignored.
- Anything left open is closed at the end of the script.

What goes in a slot

The status line always says what the slot accepts.

Value slot:

Key	Value
0–9, −	number, an integer from −32768 to 32767
A–H	variable. The same letter again: A1, A2 … A15, and back to A
X, Y	the position of the script's sprite
R and digits	RANDOM: a number from 1 to n
K and a key	KEY: 1 while it is held down, otherwise 0
T and a digit	TOUCHING: 1 when the sprite touches sprite n
Z	AT EDGE: 1 when the sprite is at the edge of the stage
P	BUTTON: the number of the last button pressed
U	COSTUME: the number of the sprite's costume (0: no costume)

Other kinds of slot:

- **Variable:** A–H, and the same letter again for A1…A15.
- **Comparison:** `<`, `=`, `>`. SPACE gives "none", and then IF only checks that the first value is not 0.
- **Sprite** (in the `WHEN` blocks): 0 the stage, 1–6 a sprite.
- **Key** (in `WHEN KEY`): RETURN and then the key.
- **Text:** RETURN, and you type up to 14 characters.
- **Variable group** (SAVE/LOAD VARS): A–H moves to the next group of eight (A-H, A1-H1 … A15-H15), and SPACE gives ALL.

The letters are read as keys, so they also work with the Greek keyboard.

The stage and the sprites

Key	What it does
arrows	move the selected sprite
1–6	select a sprite
C	next color
W, H	width, height
K	which costume it wears (none, 1, 2, ...)
E	draw its costume (a new one, if it has none)
click on the stage	puts the sprite there

A sprite without a costume is a colored rectangle.

Costumes

E opens the costume editor in place of the stage. It has a 16×16 grid with the 16 pens below it.

Key	What it does
arrows	cell
SPACE, RETURN	paints the cell
0–9, A–F	pen
.	transparent
E, TAB	done
click	on a cell: paints. On the row of pens: selects a pen

A program has up to 8 costumes, shared by all sprites. The size of a sprite with a costume is 16×16. On the stage, x is rounded to a whole byte (2 pixels in mode 0, 4 in mode 1).

All the blocks

In the "Block" column, the slots appear as in the script.

EVENTS

Block	What it does
WHEN START *sprite*	the script starts together with the program
WHEN KEY *key sprite*	starts when the key is pressed
WHEN CLICKED *sprite*	starts with a click on the sprite (or on the stage)
WHEN I RECEIVE *n sprite*	starts when message n is sent
SEND *n*	sends message n (n can also be a variable)

> A message starts **from the beginning** every script that waits for it, even if it was already running. That is why a script should not send a message to itself.

CONTROL

Block	What it does
FOREVER ... END	repeats forever
REPEAT *n* ... END	n times
IF *a comparison b* ... [ELSE ...] END	runs the lines only if the comparison holds
WAIT *n*	waits n frames (50 = one second)

WAIT UNTIL *a comparison b*	waits until the comparison holds
STOP	ends this script

MOTION

Block	What it does
GO TO *x y*	the sprite goes to x, y
CHANGE X *n*, CHANGE Y *n*	moves by n
SPEED *vx vy*	sets its speed
MOVE	one step at its speed
BOUNCE	at the edge of the stage, reverses the speed

LOOKS/SOUND

Block	What it does
COLOUR *n*	the rectangle's pen
COSTUME *n*	wears costume n (1–8. 0 changes nothing)
NEXT COSTUME	the next one, and after the last one the first again
SHOW, HIDE	
PRINT *value* AT *x y*	prints a number
STYLE *n*	the style of the PRINT and TEXT that follow: 1 bold, 2 italic, 4 underline, added together (7: all), 0 plain
NOTE *n* FOR *frames*	plays a note (MIDI number: 60 = C, 69 = A 440 Hz) and waits for it to finish
MUSIC *"name"*	plays the song *name*`.MUS` from the program's disc, behind it, with the MUSIC program (on A: or B:). It goes on after the program ends, until MUSIC's STOP (F6). It needs a memory expansion; without one: "CLOSE ANOTHER PROGRAM FIRST". A name that is not there (or empty) stops the music

VARIABLES

Block	What it does
SET *variable* TO *value*	
CHANGE *variable* BY *value*	addition
DECREASE *variable* BY *value*	subtraction
MULTIPLY *variable* BY *value*	
DIVIDE *variable* BY *value*	integer division
REMAINDER *variable* BY *value*	the remainder of the division
SAVE VARS *group* TO *"name"*	writes the variables to a text file (see below)
LOAD VARS *group* FROM *"name"*	reads them back

WINDOWS

Block	What it does
WINDOW *"title" width height*	a window in the middle of the stage, like the V-OS dialogs. A new window takes the place of the previous one and its buttons
BUTTON *"text"* AT *x y*	a system button. It is pressed with a click or with TAB and RETURN. Buttons are numbered from 0, in the order they appear

WHEN BUTTON *sprite*	starts when any button is pressed. The BUTTON value (P) says which
TEXT *"text"* AT *x y*	writes text
LABEL *button* TO *"text"*	changes the text of a button

With a window open, TEXT, BUTTON and PRINT count from the top-left corner of its inside. The window is dragged by its title, together with its buttons.

Variables

There are 128 variables, shared by all scripts, named A–H, A1–H1, ..., A15–H15. At the start they are all 0, and they hold integers from −32768 to 32767.

Saving and restoring (SAVE VARS, LOAD VARS)

- `SAVE VARS ALL TO "GAME"` writes all 128 to the file `GAME.TXT`. The `.TXT` is added when the name has no type.
- With a group (`A-H`, `A1-H1`, ...) it writes only those 8.
- The file is text, one line per variable:

```
A=12 B=0 A1=-5
```

- `LOAD VARS ALL FROM "GAME"` changes the variables the file mentions. With a group, it changes only those that are also in the group. Lines it does not understand are ignored, and if the file is missing nothing changes.
- So a file can restore all the variables or just one group of eight, e.g. a game's saved positions.

Texts in other languages

A program's texts (windows, buttons, TEXT, LABEL) can have a translation in each language of V-OS. The program shows them in the language V-OS is in when it starts. So CALC is called "ΑΡΙΘΜΟΜΗΧΑΝΗ" in Greek, "CALCULATOR" in English, "CALCULATRICE" in French.

- **EDIT > TEXT LANGUAGE (^T):** goes through the languages EN, EL, FR, ES, DE, IT and back to the program's own texts. Under the sprites, next to the name, you see the language's code (e.g. `CALC FR`).
- In a language, the text slots show and change the texts in it. A text with no translation shows its own, and when you change it, that becomes its translation. A new text that you write in a language also becomes its own.
- The translations are saved inside the `.VBS` (and the `.VBN`), all six together, about 3K.
- In a `.vbt` they are written with `LANG FR` followed by lines `"text" "translation"`.

Files

- **`NAME.VBS`:** the program as you edit it. OPEN opens it, and up to 9K is read.
- **`NAME.VBN`:** the program ready to run. RUN writes it, and it also starts from the Launcher. It is the player (`VBRUN.SYS`) together with the program. The player compiles the blocks to Z80 code when it starts, so programs run fast.

A `.VBS` can also be made from text on the PC, with `tools/mkvbs.py`. The syntax is described at the

top of the file, and examples are `blocks/examples/*.vbt` (CALC, MINES, FLAP, TICKER, VARS). CALC and MINES also have translations (`LANG`).

Limits

Lines (blocks)	240
Sprites	6
Scripts (`WHEN …`)	16
Variables	128
REPEAT in one program	64 (beyond that they share a counter)
Open blocks nested inside each other	16
Texts	32, up to 14 characters each, and in 6 languages
Costumes	8, 16×16
Buttons	64
Code	at least 4,000 bytes, even for the largest program. Whatever does not fit is ignored

Tips

- **Stage scripts:** the motion and looks blocks do nothing, and X, Y, TOUCHING and AT EDGE give 0.
- **TOUCHING:** holds only when both sprites are visible.
- **Timing:** scripts run "together". Each script gives up its turn at the end of every loop and at every WAIT. To make one script wait for another, use a variable and `WAIT UNTIL` (see MINES).
- **Ready-made examples:** CALC, MINES, FLAP and TICKER open in Studio (OPEN).

6. Blocks on the PC: the web editor and `vbc.py`

Besides Studio, which runs on the CPC, there is also a block editor for the PC. The language is the same, with a few extra blocks: lists, noise, sprite size, colored text.

The editor

Open `blocks/editor.html` in a browser. It is based on Blockly and comes in Greek or English.

- **Categories:** events, control, motion, looks, text, operators, sensing, variables, lists and windows.
- **Sprites:** rectangles or costumes. They are defined in a table, and the stage shows them in the CPC's colors. They can be dragged with the mouse.
- **Files:** it opens and saves `.vbp` files. They are JSON with the whole program, so they can also be written by hand.

Compiling

```
python3 blocks/vbc.py pong.vbp --dsk build/vos.dsk
```

`vbc.py` writes `build/NAME.ASM`, with a comment for each block. It assembles it with RASM into `NAME.VBN`, with a relocation table so that it runs alongside other programs, and with `--dsk` it puts it on the disk.

Option	What it does
`-o build`	output folder
`--asm-only`	only the `.ASM`
`--dsk build/vos.dsk`	also onto the disk

The `.vbp` format

The full description is at the top of `blocks/vbc.py`: events, commands, values, costumes, icon. In short:

```
{"name": "PONG", "mode": 0, "background": 0,
 "sprites": [{"name": "ball", "x": 90, "y": 100, "w": 6, "h": 6,
              "colour": 8, "vx": 2, "vy": 2, "visible": true}],
 "variables": {"score": 0},
 "scripts": [{"sprite": "ball", "when": {"event": "start"},
              "do": [{"block": "forever", "do": [{"block": "move"}, {"block": "bounce"}
```

- **Variables and lists:** they have names, not letters. Lists have a fixed maximum length.
- **Icon:** `"icon"` gives an icon for the Launcher.
- **A program with many similar parts:** a Python script can write it. See `tools/mkbricks.py`, which writes `bricks.vbp` with 32 bricks.

Examples

In `blocks/examples/`:

File	Program
`pong.vbp`	PONG

`catch.vbp`	CATCH
`starcat.vbp`	STARCAT: costumes, a tune from a list
`counter.vbp`	COUNTER: a window and buttons
`bricks.vbp`	BRICKS (written by `tools/mkbricks.py`)

The `*.vbt` files are Studio programs as text, for `tools/mkvbs.py`. See chapter 5.

Studio or web editor?

	Studio	**web editor**
Where	on the CPC	on the PC
File	`.VBS`	`.vbp`
Variables	128, A–H15	with names
Lists, noise, sprite size	no	yes
SAVE/LOAD VARS	yes	no
Text style, `.MUS` music	yes	yes
The program	needs all the memory	small, runs alongside others

7. The V-OS SDK: programs in assembly

With this SDK you write your own programs in Z80 assembly. They run inside V-OS like DEMO or CLOCK: they appear in the Launcher, the Desktop and the Commander, and they have their own menu, buttons, dialogs, sound and files. They also run alongside other programs.

Quick start

```
make new-app NAME=hello
```

This creates the folder `myapps/hello/` from `sdk/template/`. It has three files:

- `hello.asm`: the program. The arrows move a box, there is a BEEP button, a seconds counter, and its own menu (ABOUT, BACK) next to VIEW and HELP.
- `texts.txt`: its texts, in English and Greek.
- `icon.txt`: its icon, 16×16, for the Launcher.

```
make app NAME=hello
```

This builds `build/HELLO.BIN` and puts it on the programs' disk, `build/apps.dsk`. `make run` puts that in B:; in the Launcher, TAB shows B:.

```
make run
```

This opens the disk in Caprice32. HELLO is in the Launcher, on the last page.

- Plain `make` builds all the programs in `myapps/` and puts them on the disk, together with V-OS.
- The name has 1–8 lowercase letters or digits and starts with a letter. It cannot be the name of a V-OS program (e.g. demo).
- You need RASM, Python 3 and iDSK, as for all of V-OS.
- In Caprice32 you run the program yourself, from the Launcher.

A bigger program:

- Put more `.asm` files in its folder (e.g. `art.asm` with sprites) and bring them in with `include "art.asm"`. The build finds them, and rebuilds the program when they change.
- Whatever you put in the subfolder `disc/` (e.g. `disc/LEVEL1.DAT`) goes onto the disk as it is, for `VOS_LOAD`.
- With a subfolder `owndisc/`, the program also gets a disk of its own, `build/<name>.dsk`, with the `.BIN` and the files of `owndisc/`. It stays off `apps.dsk`. You put it in B: and run it from the Launcher (TAB, the other drive). Which drive it was started from, so that it loads its files from there, is in `VOS_SELF`: `db drive`, then its name (11 bytes).
- **A file to open (version 14):** when RETURN in the Desktop or the Commander opens a file in your program, `VOS_OPEN` (`#B000`) holds `db 1, drive, 11-byte name` (0: none). Read it in `init`, with A = 0 and also with A = 2 (the program was already running), then write 0 there. The types and their programs are in the kernel's `ft_open` table (`src/kernel/files_ext.asm`).
- A program with a relocation table (like those of the SDK) runs beside others when it is up to 12K; a bigger one (up to 16K) runs on its own without a memory expansion, like one without a table (MARKDOWN, READER). For the rest, load files into the data bank.

How a program works

A program is a `.BIN`, built for `VOS_APP_ADDR` (`#4000`). It starts like this:

```
        include "api/vos_api.inc"
        org VOS_APP_ADDR
        jp init                 ; +0
        jp event                ; +3  HL -> the event
        jp redraw               ; +6
        db "VA", VOS_API_VERSION
        include "app_icon.asm"  ; +12 the icon (optional)
```

- **`init`** sets mode, palette, menu and buttons, turns on `VE_FRAME` if needed, and draws everything.
  - ○ A = 0: a fresh start.
  - ○ A = 2: the program returns from a view or another program. Its state is as it left it; it sets the above again and draws.
- **`event`** is called for each event and returns immediately. A program never waits on its own (no wait loop). For motion or time, `VOS_TICKS` with A = 1, and then a `VE_FRAME` arrives 50 times a second.
- **`redraw`** redraws everything, e.g. when the language changes or the desk accessory closes.
- **Exit:** `VOS_EXIT`, which returns to the view that started it.

The order inside `event`

```
event:  ld a,(hl)
        cp VE_FRAME : jp z,frame        ; the frames
        push hl
        call VOS_UI_EVENT               ; menus, buttons, CONTROL+letter
        pop hl
        or a : jr nz,command            ; A = the command
        call VOS_MENU_OPEN : ret nz     ; menu open: the keys are its own
        ...                             ; your own keys and clicks
command: ...
        jp VOS_COMMAND                  ; whatever is not yours (VIEW, HELP)
```

While a menu is open, do not draw. Otherwise you will draw over it. On `VE_FRAME`, update the state, but draw only if `VOS_MENU_OPEN` gives Z. Dialogs (`VOS_DIALOG`, `VOS_INPUT`, `VOS_FILE_PICK`) have their own loop, so no event arrives while they are open.

The events

`HL` → 8 bytes: `db type, code, char, mods; dw x, y`.

type	what	code / char / x, y
`VE_KEY_DOWN` 1, `VE_KEY_REPEAT` 3, `VE_KEY_UP` 2	key	code = key number (`VK_*`, keyboard row·8 + bit), char = the character in the pack's language
`VE_PTR_DOWN` 5, `VE_PTR_UP` 6, `VE_PTR_MOVE` 4	pointer	x, y in pixels of the mode
`VE_FRAME` 8	frame (with `VOS_TICKS`)	—
`VE_LANG` 7	the keyboard language changed	—

Three kinds of program

- **With a relocation table (the SDK default):** `make app` builds it twice, at `#4000` and at `#4100`. `tools/mkreloc.py` adds the table from the difference.
  - It runs alongside other programs, in the 12K from `#5000`, and continues where it stopped (F6 shows the running programs).
  - Its data must be inside its own bytes (labels, `ds`), never at fixed addresses.
  - If `mkreloc.py` gives an error, an address is used somewhere in a way that does not survive relocation. E.g. `ld a,label & 255`, or only the low byte of an address.
- **Without a table:** it takes the whole bank (up to 16K, code and data), but runs only on its own, unless there is a memory expansion. This needs your own rule in the Makefile (see `STUDIO.BIN`).
- **Desk accessory:** up to 4K, with a table. It opens with F5 on top of anything, and `init` gets A = 1. See `apps/clock/clock.asm` and the description in `src/api/vos_api.inc`.

Rules

- **Registers:** calls corrupt all registers except IX. Arguments go in A, BC, DE, HL (IX for `VOS_RECT` and `VOS_BLIT`). Results return in A or HL. On failure, carry is set and A = error number, which `VOS_ERROR` displays.
- **Memory:** do not page another bank in at `#4000`, because the program will vanish from under you. For 16K more there is the data bank (`VOS_BANK_GET`, `VOS_BANK_PUT`, `VOS_DATA`). It is shared with views and other programs, and keeps its data only while the program is running. With a memory expansion (API 21) each program gets one of its own, which keeps what is in it until the program ends, whatever runs meanwhile; `VX_DATA` says which (A = 1: its own).
- **Coordinates:** x in pixels of the mode (mode 0: 0–191, mode 1: 0–383, mode 2: 0–767), y from 0 to 258. The menu bar takes y 0–11.
- **RASM:** labels are not case-sensitive (`BOX` and `box` are the same). Hex is written with `#` and not with `&`.

The data formats

- **Menu bar** (`VOS_SET_BAR`): `db count; dw menu, ...`. The shared VIEW and HELP come from `VOS_MENU_VIEW` and `VOS_MENU_HELP`.
- **Menu:** `db title, count` and for each item `db text, command, letter, 0`.
  - text: `VOS_SEPARATOR` for a line
  - command: 1.., what `VOS_UI_EVENT` returns
  - letter: CONTROL+letter shortcut, or 0
- **Buttons** (`VOS_SET_WIDGETS`, then `VOS_DRAW_WIDGETS`): `db count`, and for each button `dw x, y; db width in characters, text, command, 0`.
- **Dialog** (`VOS_DIALOG` → A = button, 0 the first, ESC gives the last): `db title, text, number of buttons (1-3), button, ...`. For buttons, `VM_OK`, `VM_YES` and `VM_NO`.
- **Texts** (`VOS_TEXTS`): made by `tools/mktexts.py` from `texts.txt`, with lines `NAME | English | Greek`. Both columns are written in UTF-8 and become bytes of the V-OS character set (`tools/voscs.py`), which holds Greek and the Latin letters of French, Spanish, German and Italian.

    - It gives `T_NAME` (`VT_FIRST` = `#C0` and up) and the table `texts:`. Up to 64 texts.
    - A `|` with no spaces around it starts a new line in dialogs.
    - Texts go wherever a "text" is asked for (menus, buttons, dialogs). With `VOS_MSG` you get their address for `VOS_PRINT`.

- **Palette** (`VOS_PALETTE`): 17 firmware colors (0–26): 16 pens and the border. In mode 1, pens 0–3 count.
- **Rectangle** (`VOS_RECT`, IX →): `dw x, y, w, h; db pen`.
- **Icon:** 16 lines of 16 characters in `icon.txt`. Each character is a pen in hex (mode 0, Launcher palette) or `.` for transparent. The Launcher reads it from the disc: while the program runs, the 256 bytes after "IC" are free for buffers (PAINT keeps its rows there).
- **File names** (`VOS_LOAD`, `VOS_SAVE`, `VOS_DELETE`, `VOS_RUN`): 11 bytes, `"NAME EXT"`. Drive 0 = A:, 1 = B:.

The API (version 22)

Call	Input → output	What it does
`VOS_EXIT`	—	end, back to the view that started it
`VOS_TICKS`	A = 1 / 0 (+ `VOS_PTR_KEYS`)	`VE_FRAME` 50 times a second, or none. With `VOS_PTR_KEYS` (2) the arrows and COPY also move the pointer and click
`VOS_FRAMES`	→ HL (and A bits 16–23)	frames since startup
`VOS_RANDOM`	→ A	0–255
`VOS_COMMAND`	A = command	the shared commands of VIEW and HELP (6–19). Number your own commands from `VC_APP` (32) up, or a VIEW or HELP item runs one of yours
`VOS_ERROR`	A = error	error dialog
`VOS_MODE`	A = 0 / 1 / 2	mode 0 (192×259, 16 pens), 1 (384×259, 4 pens) or 2 (768×259, 2 pens, 96 text columns; version 25). In mode 2 menus and dialogs draw in pen 1 on pen 0: pen 0 white and pen 1 black give black text on white paper (NOTEPAD, MARKDOWN, READER, SHEET with ^M)
`VOS_PALETTE`	HL → 17 colors	
`VOS_CLS`	A = pen	clears the screen
`VOS_PLOT`	HL = x, DE = y, A = pen	one pixel
`VOS_RECT`	IX → rectangle	filled rectangle
`VOS_COLOURS`	A = text pen (+ `VOS_BOLD`, `VOS_ITALIC`, `VOS_UNDERLINE`), C = background	for `VOS_PRINT` and `VOS_CHAR`. The styles are made from the same font, for every language
`VOS_PRINT`	HL = x, DE = y, BC → text, 0	
`VOS_CHAR`	HL = x, DE = y, A = character	
`VOS_MSG`	A = text → HL	the address of a text (`T_*` or a system one)
`VOS_SET_BAR`, `VOS_DRAW_BAR`	HL → menu bar	

VOS_UI_EVENT	(the current event) → A	menu, button or shortcut command, 0 for none
VOS_DIALOG	HL → dialog → A	
VOS_INPUT	A = title, HL → text, C = maximum → carry if cancelled	text input (needs C + 2 bytes)
VOS_SET_WIDGETS, VOS_DRAW_WIDGETS	HL → buttons	
VOS_MENU_VIEW, VOS_MENU_HELP	→ HL	the shared menus
VOS_CATALOG	A = drive → A = files, HL → list	16 bytes each: 11 the name, at +12 the size in K, at +13 1 for a folder (drive 2). With A = drive + VOS_CAT_EXT and HL → "EXT", only the files with that extension
VOS_FILE_PICK	A = title → A = file, carry if cancelled	picker dialog, after VOS_CATALOG
VOS_LOAD	A = drive, HL → name, DE = where → BC = length	
VOS_SAVE	A = drive, HL → name, DE = from, BC = length	binary file, replacing the old one. With A + VOS_LOAD_IX (#40, version 20) its header says it loads at IX, not at DE (a .SCR: IX = #C000)
VOS_DELETE	A = drive, HL → name	
VOS_TEXTS	HL → text table	the program's texts
VOS_MENU_OPEN	→ NZ if a menu is open	
VOS_KEY_HELD	A = key → NZ while pressed	for games
VOS_BLIT	IX → `dw x, y; db bytes, lines; dw source; db source stride`	image: pairs (mask, pixels)
VOS_SOUND	A = channel 0–2, HL = period (bit 15: noise), C = volume 0–15, B = frames	a note; 440 Hz is period 142
VOS_SPLIT	HL → `dw y1, y2; db 8 colors` (0 = turns off)	mixed screen: mode 0 between y1 and y2
VOS_RUN	A = drive, HL → name	runs another program, and then this one starts again
VOS_KEEP	→ HL → 16 bytes	memory that stays between programs
VOS_BANK_GET, VOS_BANK_PUT	HL = from, DE = to, BC = bytes	copy from and to the data bank
VOS_BACKGROUND	A = 1 / 0	keeps running or freezes when behind another
VOS_DRAG	HL = x, DE = y, BC = w, A = h → carry: new HL, DE	dragging a window by its title
VOS_FIT	same → HL, DE	brings a window inside the screen
VOS_EXT	A = a VX_* call, BC, DE, HL as it wants → A, HL, C, DE, carry	the calls after the table (version 15), below

Drives: 0 = A:, 1 = B:, 2 = DISK, the SD card of an M4 board (version 28), in the folder the Desktop or the Commander last went into. File calls and VX_FREE work there as on a disc (8.3 names; without an M4 board: carry, A = 35). Databases (VX_DB) work there too (version 29). From version 29, VOS_CAT_EXT keeps the folders, and VOS_FILE_PICK goes into a folder chosen (or ..) and lists it

as the last VOS_CATALOG asked, so what it returns is always a file, in the folder now. VX_DRIVE C = drive → A = the next one (A: → B: → DISK with an M4 board → A:), for an OTHER DRIVE command; src/api/vos_drive.inc has vos_next_drive (A: ↔ B: on an older V-OS) and vos_drive_text (A:, B:, DISK:).

Network: src/api/vos_net.inc (include it after vos_api.inc) gives TCP connections through the M4 board for your own protocols, as many at once as you like (one name looked up at a time). Each connection is NC_SIZE bytes of yours at IX, all 0 at first. net_open HL → "name", "name:port" or "1.2.3.4:port", DE = the port when it has none → looked up and connected (carry, A = NE_NO_M4 or NE_FAILED: not). net_poll at each VE_FRAME, again while it does not give NE_NONE → A = NE_CONNECTED, NE_LINE (HL → a line without its CR LF, 0 at its end, B = its length; a long one comes in pieces with net_long = 1), NE_DATA (with NC_RAW: HL → bytes, BC = how many), NE_CLOSED, NE_FAILED, NE_NO_M4. net_say HL → a line: sent with CR LF; net_send HL, BC (up to 240); net_ready → carry while the board is still sending; net_close; net_reply HL → a line of an SMTP or FTP reply → carry if it is not the last, else DE = its number. Examples: MAIL (myapps/mail: POP3 and SMTP) and FTP (myapps/ftp: two connections, PASV). No SSL/TLS. To test without a board: tools/m4net.py (an M4 board and servers in Python).

The VOS_EXT calls: VX_VERSION → A = API, HL = V-OS version (BCD). VX_FREE C = drive → A = free K. Unknown: carry VX_CLIP_PUT HL = text, BC = bytes → the clipboard (up to 1K). VX_CLIP_GET DE = where, BC = room → HL = size (version 16). VX_M4 HL → a command for the M4 board (db size, dw command, data), DE → the response, BC = bytes; carry: no M4 board (version 17); HL = 0: BC bytes of the M4 ROM's sock_info to DE, 16 a socket (its status, its last command, the bytes received, an ip and a port; socket 0 is C_NETHOSTIP's, its ip the name looked up) (version 32; TELNET waits on it). VX_MEM → HL = the RAM in K, DE = the K free for programs (version 18). VX_BEHIND DE → db drive, 11-byte name of a file: its program (by type) starts behind this one, with init A = 3 and the file in VOS_OPEN (version 19; MUSIC knows it). VX_DATA → A = 1 if the data bank is the program's own (memory expansion), 0 if it is the shared bank 4 (version 21; NOTEPAD keeps text up to 16K this way). VX_DB C = a DB_* call, DE, HL → a database (version 22, below). VX_UTF8 HL → text, BC = bytes, E = 0 in the program's memory or VOS_DATA in its data bank: if it is UTF-8 (from a PC or the web), it becomes the V-OS character set there → BC = bytes now, A = 1; A = 0: it was not UTF-8 and stays as it was (version 23; NOTEPAD and READER do this). VX_CALC C = a CALC_* call, DE → a block, HL → text → formulas and numbers (version 24, below). VX_LANG → HL = the code of the language V-OS shows ("EN", "EL", "FR"...; L the first letter) (version 26; VBRUN shows the translations of Studio's programs this way). VX_IRQ HL = a routine of the program → the kernel calls it once a frame from the interrupt, in front or behind, with its bank there and interrupts off; 0 stops it, and so does the program's end; carry: not for an accessory or a program alone in bank 5 (version 27; MUSIC plays its songs this way, so they do not stop while another program loads or draws). It must be short, keep IY, EXX and AF', write only its own memory and the AY, and call nothing of V-OS. VX_PRINT HL → text, BC = bytes, E = 0 in the program's memory or VOS_DATA in its data bank → to the printer (the CPC's Centronics port, 7 bits), as it is: 13, 10 end a line, 12 starts a new page; the letters over 127 go out as plain ASCII (accents left out, Greek as Greeklish: Θ 8, Ξ 3, Ψ 4); carry, A = 41: the printer stayed busy for two seconds (none, off line, no paper) (version 30; NOTEPAD, SHEET and DATABASE print with ^P). After 60 lines a new page begins by itself; BC = 0 with HL → a file's 11-byte name (as VOS_CATALOG gives it) starts a printout whose pages begin with a heading, NOTES.TXT PAGE 2; HL = 0: no headings (version 31). VX_VIEW HL → dw x, y, w, h: from then on the program draws with (0, 0) at (x, y) and nothing outside that rectangle (cut to the screen): VOS_CLS fills it, VOS_RECT is cut to it, a point, a character (each of VOS_PRINT) or a VOS_BLIT image is drawn only if all of it is inside; menus, dialogs and the bar are not affected; HL = 0: none; set it again after VOS_MODE and in redraw (it ends when another program comes to the front); carry: no memory expansion (version 33; the first step towards programs in windows, docs/WINDOWS.md). VX_WINDOW HL → dw w, h: the program in a window on the desk, its inside w × h (w a multiple of 8) as its view and its pointer events with x, y from it; call it first in init, each time (A = 2 too: that is how the windows behind are drawn again, through their init); its title moves it, X ends it, + or F7 make it

full screen and back, a click on another brings that one to the front with its bar; mode 1, the theme's colours; carry: no memory expansion or no `WM.SYS` on A: (full screen as before; version 34). From version 35 `VX_WINDOW` writes back at HL the size its inside has now (full screen: the screen under the bar), so init lays out what it shows for it; the bottom right corner resizes it (and init is called again with A = 2); with `VOS_BACKGROUND` the program draws in its window from behind too, while none covers it (`VOS_CLS`, `PLOT`, `RECT`, `COLOURS`, `PRINT`, `CHAR`, `BLIT`; not `VOS_MSG`). Examples: `apps/clock`, `myapps/sysinfo`. From version 36 HL points to dw w, h, x0, y0: the window shows the part of the program's screen from x0, y0 (0, 0: as before), so a program made for the whole screen draws as it did, its widgets and events going with it (e.g. `dw 376, 232, 0, 12` in `myapps/telnet`); a program with version 36 in its header always gives all four. One that calls it outside init draws its bar again after it (`VOS_DRAW_BAR`). From V-OS 0.49 a window behind that has a copy of itself (taken while nothing covered it) is copied back instead: its init is not called then, so a program must not count on being called to draw while behind; one that goes on in the background (`VOS_BACKGROUND`) draws as before while uncovered. The Desktop's files window and drive icons are among the windows too (0.50): up to eight windows in all. From 0.51 `VOS_PALETTE` does nothing while the program is in front in a window (full screen it works); a window's X sends the program CONTROL+Q, so its BACK may ask first; a program behind that nothing covers has `VOS_EXT` too (TELNET receives there), and a `VOS_EXT` that did nothing while covered leaves your buffers as they were: mark them first, as TELNET does. With w = 0 (V-OS 0.48) the program leaves the windows: its window goes (carry), and it sets its mode and palette and draws full screen, as for a mode 2 page; asking again later brings the window back where it was, as large as it was. The size written back lets a program lay itself out for its window: NOTEPAD, MARKDOWN, READER and SHEET (`set_mode`) take w / 8 columns and as many rows as fit, again at each resize, and leave the windows for their page (^M).

Databases (version 22)

`VOS_EXT` with A = `VX_DB` and C = a `DB_*` call works on a database: a file `NAME.VDB` of 1K pages, as big as the disc allows (up to 255K). It holds up to 8 tables; a table has up to 10 fields, text (up to 255 characters) or a number (2 bytes), and may have a key field (text up to 64 characters, or a number) with an index, so that FIND is fast. A record is its fields one after the other; its id (rid) is a 16-bit number. One database is open at a time, for the program that opened it. Each call writes its changes to the disc before it returns. The kernel loads the engine, `DB.SYS`, from drive A: the first time.

Call	In → out
`DB_CREATE`, `DB_OPEN`	DE → db drive, 11-byte name (`"STREETS VDB"`); OPEN → A = 1: Latin-1 (of V-OS 0.15)
`DB_CLOSE`	
`DB_TABLE`	DE → name (8), fields, key field (0: none), then each field: name (8), type, length → A = table
`DB_USE`	E = the table the calls below use
`DB_INFO`	E = table, HL → 124 bytes: its descriptor → A = tables
`DB_ADD`	HL → record → DE = rid
`DB_GET`, `DB_PUT`	DE = rid, HL → the record
`DB_DEL`	DE = rid
`DB_FIND`	HL → key as in a record (room for a record), E = bytes that count (0: all) → DE = the first record with that key or after it
`DB_FIRST`, `DB_LAST`, `DB_NEXT`, `DB_PREV`	HL → room for the record, or 0 → DE = rid, in key order; carry at the end
`DB_COUNT`	→ HL = records

Errors: carry and A (DB_ERR_*, and the disc's errors), which VOS_ERROR shows. Text is kept as it is typed, in the V-OS character set, the same in every language: one database can hold Greek, French and German together. Only a database of V-OS 0.15 with Latin letters (Latin-1) is different: DB_OPEN then gives A = 1; show DB_ERR_ALPHABET with VOS_ERROR. Sorting and FIND compare bytes: in a DB_TEXT key capitals and small letters, and accented ones, are different. A DB_ALPHA field (type 2, from version 23) is text like DB_TEXT, but as a key it is sorted and found without case or accents: ΕΛΕΝΗ, Ελένη and ελένη, or Élodie and ELODIE, are the same; the record keeps the text as it was typed. Examples are STREETS (myapps/streets), a street index, and DATABASE (myapps/database), a whole database program with tables, a list and a form; tools/vdb.py shows and checks a .VDB file on the PC.

Formulas (version 24)

VOS_EXT with A = VX_CALC and C = a CALC_* call works out formulas, as a spreadsheet does. The kernel loads the engine, CALC.SYS, from drive A: the first time. A number is 32 bits, in hundredths (1234 is 12.34), from -21474836.46 to 21474836.46. A formula has + - * /, brackets, numbers (12, 3.5, .25; a third decimal rounds), cells (A1 to Z255) and the functions SUM, AVG, MIN, MAX, COUNT (of values, cells and ranges such as B2:D9, between , or ;), ABS(x), INT(x) (down to a whole number) and ROUND(x) or ROUND(x, 0–2), in capitals or not. Products and quotients are rounded to the hundredth.

DE → the program's block, CB_SIZE (10) bytes: CB_RESULT (a number, 4 bytes), CB_COLS, CB_ROWS, CB_GRID (dw), CB_BANK, CB_AT. The cells are a grid of the program's: CB_COLS × CB_ROWS numbers of 4 bytes, row after row, at CB_GRID, in its data bank when CB_BANK = VOS_DATA. A cell CALC_NONE has no number (empty, or text: 0 in + - * /, left out by the functions); CALC_BAD has an error.

Call	In → out
CALC_EVAL	HL → formula, 0 (= first or not) → CB_RESULT and DE:HL (DE high)
CALC_NUMBER	HL → text: only a number? → as CALC_EVAL
CALC_TEXT	CB_RESULT → text at HL (13 bytes of room: -12.5), A = its length
CALC_CELL	HL → text: a cell (b12) at its start? → D = column, E = row (from 0), A = its characters

Errors: carry and A (CALC_ERR_*: 30 a wrong formula, 31 division by zero, 32 a number too big, 33 a cell outside the grid, 34 a cell it uses has an error, 29 CALC.SYS is not on A:), which VOS_ERROR shows; then CB_RESULT = CALC_BAD and CB_AT = where in the text. The example is SHEET (myapps/sheet), a spreadsheet that keeps its values in its data bank and works the formulas out again after each change.

Full details, e.g. on the background, the accessory and VOS_SPLIT, are in src/api/vos_api.inc. For examples, see apps/demo/demo.asm (mode 0, motion), apps/clock/clock.asm (accessory) and apps/mixed/mixed.asm (mixed screen). A whole game with sprites, a stage from a file and sound is myapps/vrouves/.

An AI assistant writing a program should read AI.md: the same rules, shorter, with the traps of RASM and of the data bank.

Testing

make test-sdk builds the template as TEMPLATE.BIN and opens it from the Launcher in Caprice32, with key presses defined by src/kernel/test_script.asm (script 15). It takes screenshots in build/shots: its screen, the box after it moves, its menu, the ABOUT dialog, and the return to the Launcher.

For your own automated test, copy script 15 and the test-sdk target of the Makefile.

8. Developing V-OS

For those who work on V-OS itself. Programs *for* V-OS are described in the SDK.

What you need

- RASM, the assembler
- iDSK, for the disk images
- Python 3, for the tools. `make docs` also needs `markdown` and `pymupdf`.
- GNU make 4.3 or newer
- for testing, Caprice32 (snap `caprice32`)

Build

Command	What it does
`make`	`build/vos.dsk` (A:): the kernel, the views, the languages, DEMO, MIXED, STUDIO, CLOCK and the `.VBS` sources; `build/apps.dsk` (B:): the other programs and `myapps/*`; and the disc of each program with `owndisc/`. If a file does not fit on a real disc (180K), it stops with an error
`make run`	opens the disk in Caprice32 (`RUN"VOS"`)
`make shot`	starts without a window and takes a screenshot
`make docs`	the manual as PDFs: `docs/V-OS.pdf` (Greek) and `docs/V-OS-en.pdf`, `-fr`, `-es`, `-de`, `-it`
`make windows`	`build/V-OS-windows.zip`: `V-OS.exe` for Windows, with CPCEMU 3.0.2 (CPC 6128, 576K, AMX mouse, M4 board), `vos.dsk` in A:, `apps.dsk` in B: and the programs' own discs beside them. V-OS starts by itself. Needs WSL and the portable CPCEMU zip (`CPCEMU_ZIP=...`)
`make linux`	`build/V-OS-linux.tar.gz`: the `V-OS` script for Linux (x86_64), with CPCEMU 2.5 (3.0.2 is not out for Linux) and the same settings and discs. Needs SDL2 on the system and CPCEMU's Linux tar.gz (`CPCEMU_TGZ=...`)
`make songs`	the songs of Arkos Tracker 3 as `.MUS` files in `myapps/music/owndisc` (`tools/mkmusic.py`; needs Arkos Tracker 3, `ARKOS=...`)
`make new-app NAME=x`, `make app NAME=x`	your own program (SDK)
`make clean`	deletes `build/`

Repo layout

Folder	Contents
`src/vos.asm`	the root of the build: loader and kernel, writing to the disk
`src/boot/`	the loader that AMSDOS runs at `#4000`, RAM expansion detection
`src/kernel/`	vectors, interrupts, file system, views, API (`api.asm`), settings, kernel extension (`ext.asm`), test scripts (`test_script.asm`)
`src/drivers/`	Gate Array, CRTC, keyboard, AMX mouse, FDC, AY
`src/gfx/`	overscan graphics, text, mixed screen
`src/ui/`	menus, buttons, dialogs, text input, file picker

`src/shell/`	the views as overlays: Desktop, Launcher, Commander, Setup
`src/api/` `vos_api.inc`	the API for programs
`lang/`	the languages (texts, keyboards, fonts)
`apps/`	the programs in assembly: DEMO, MIXED, STUDIO, CLOCK, and their icons
`blocks/`	the web editor, `vbc.py`, `runtime.asm`, the Studio player (`player.asm`), examples
`sdk/`	the SDK: guide and program template
`tools/`	tools: fonts, languages, texts, icons, relocation, `.VBS` from text, screenshots, PDF
`docs/`	this manual in Greek (`manual/`) and in English, French, Spanish, German and Italian (`manual/en/`, `fr/`, `es/`, `de/`, `it/`), the PDFs, the requirements (`REQUIREMENTS.md`)

Memory, in brief

Area	What it holds
`#0000-#3DFF`	the kernel. In page zero, after the restart vectors, the API table at `#0108`
up to `#4000`	the kernel stack
`#4000-#7FFF`	the bank "window"
`#8000-#FFFF`	the 32K overscan screen. In its gaps, kernel buffers

The banks that go into the window:

Bank	What it holds
4	file and load buffer, and the data bank of the running program
5	the view at `#4000-#4FFF`, and from `#5000` the programs with a relocation table (or one program without a table in the whole bank)
6	whatever the menus and dialogs hide, and from `#7000` the desk accessory
7	the language pack, the Commander lists, the English texts, and from `#7000` the kernel extension

With a memory expansion, its banks take programs. The details, address by address, are in `docs/REQUIREMENTS.md` (§4.2 and in each "Υλοποίηση Μ..." (implementation) section).

Conventions

- **RASM:** hex is written `#C000` and not `&C000`, because RASM reads `&` as AND. Labels are not case-sensitive.
- **Comments:** in English, short, as in the existing code. Each file starts with a block that says what it does.
- **Documentation:** in Greek. `MILESTONES.md` (English), `MILESTONES.el.md` (Greek) and `docs/REQUIREMENTS.md` have a section for each milestone. The manual is written in Greek and translated into the other five languages of V-OS.
- **Texts:** the system texts are in `lang/*/messages.txt`, and the programs' texts in `texts.txt` (for `tools/mktexts.py`).

Testing

The tests run V-OS in Caprice32 without a window and take screenshots in `build/shots/<delay>/`.

How they work:

- Caprice32 without a window cannot press arrows, COPY or f keys. That is why the key presses are made by a special build: a `-DTEST_SCRIPT=n` puts script n into `src/kernel/`

`test_script.asm`, a table of keys per frame.

- Each `make test-…` target builds a disk with its script and takes a screenshot at each delay in `…_SHOTS`.
- The delay is measured in "Z" presses: about 2.5–3.5 frames each. The mapping is not fixed, so you catch a moment with two or three delays around it.
- Once V-OS writes to the disk, Caprice32 does not close by itself, and the test ends with a timeout. This is expected in test-vars, test-costume and others.
- You look at the screenshots yourself, e.g. combined into one image with `ffmpeg … hstack/vstack`.

Target	What it tests
`test-input`	keys, Studio: blocks, files, running
`test-calc`	CALC with clicks, then `CALC.VBS` in Studio and its title in English and French (^T)
`test-costume`	costume in Studio, running, reopening
`test-acc`	the clock as a desk accessory
`test-multi`, `test-switch`, `test-expand`, `test-behind`	several programs, F6, memory expansion, background
`test-drag`	dragging windows
`test-langs`, `test-lang`	the languages, saving the language
`test-mines`, `test-bricks`	the games
`test-vars`	the 128 variables, SAVE/LOAD VARS
`test-sdk`	the SDK template
`test-apps`	the two discs as given out, `vos.dsk` (with the test keys) in A: and `apps.dsk` in B:: the Launcher shows B: and PONG runs from there
`test-vrouves`	VROUVES from its disc in B:, through the Launcher (TAB): building, wave 1, the funds, HIGH SCORES
`test-notepad`	NOTEPAD: typing, DEL, the save dialog, OPEN with only `.TXT` files; then ^M: the page (mode 2)
`test-notepad-big`	NOTEPAD's text window in a Z80 emulator (`tools/notepad_check.py`): a 12K file with its own data bank, moving, typing and SAVE, text up to 16K, and 4K on a 6128
`test-data`	(576K) NOTEPAD with a 15K text down to line 33, MARKDOWN meanwhile, then NOTEPAD back up to the start: the lines from its own data bank
`test-paint`	PAINT: line, box, fill, the menus, then the layers (hiding one), OTHER DRIVE and EXPORT .SCR to B: ("SAVED AS .SCR")
`test-paint-big`	PAINT on the whole screen (576K): lines, the window moved by its bars (a click and a drag), its place, saved as `B:NONAME.SCR` and opened again
`test-scr`	PAINT's EXPORT .SCR (its code in a Z80 emulator, `tools/scr_check.py`) loaded in BASIC from a disc, with the palette; every pixel of the screen checked
`test-about`	HELP > ABOUT V-OS
`test-music`	MUSIC from `music.dsk` in B:: a song plays, and goes on behind the Launcher
`test-markdown`	MARKDOWN from `markdown.dsk` in B:: `README.MD` opened, the page with ^P, a screen down, back to the text; then ^M: the page (mode 2)
`test-open`	opening by type in the Commander: `README.MD` in MARKDOWN from B:, a `.TXT` in NOTEPAD from A:

`test-sysinfo`	SYSINFO from the Commander: version, API, memory, free K on A: and B:, ^R, ^Q. With `SYSINFO_RAM=576`, with a 512K expansion
`test-clip`	the clipboard: selecting with SHIFT, ^C and ^V in NOTEPAD, ^A, ^X, ^V, then ^V in MARKDOWN
`test-webget`	by hand, in the CPCEMU of `make linux` (M4 board; needs a screen and the internet): WEBGET downloads `example.com/index.html`. Close CPCEMU with F12 when SAVED shows, and `INDEX.HTM` is looked for on the disc
`test-reader`	READER from `reader.dsk` in B: opens `INDEX.MD`, TAB selects a link, RETURN opens `ABOUT.MD`, DEL goes back, and an `https://` link gives a message; then ^M: the page (mode 2)
`test-db`	the database engine in a Z80 emulator (`tools/db_check.py`): thousands of random calls on four tables, checked against a model, the file page by page (`tools/vdb.py`), until the disc is full
`test-db-disc`	`DB.SYS` with the kernel's file system on a DSK image (`tools/db_disc_check.py`)
`test-streets`	STREETS from `streets.dsk` in B:: a new `STREETS.VDB` with 24 streets, finding, a new street
`test-database-z80`	DATABASE in a Z80 emulator, with the database in Python and the screen as text: two tables, 47 records from the form, the order without case or accents, finding, a change, a deletion, scrolling
`test-database`	DATABASE in Caprice32: a new database on B:, a table, three records from the form, the list, finding, a change
`test-calc-sys`	the formula engine `CALC.SYS` in a Z80 emulator (`tools/calc_check.py`): fixed formulas and 40,000 random ones, numbers and texts, checked against a model in Python
`test-sheet-z80`	SHEET in a Z80 emulator with `CALC.SYS` itself and the screen as text: values, `#ERR`, cells that use each other, scrolling, SAVE and OPEN, a UTF-8 file, `BUDGET.VSH`, a full sheet; then ^M: the page (mode 2)
`test-sheet`	SHEET in Caprice32: numbers and a formula typed, `CALC.SYS` from A:, the sheet saved on B:; then ^M: the page (mode 2)
`test-kbds`	KEYBOARDS: ALL SIX in Setup, then in NOTEPAD the `:` and Z keys with each keyboard: `:z`, `:ζ`, `mw`, `ñz`, `öy`, `òz`
`test-charset`, `test-reader-utf8`	UTF-8 files in Greek, French, Spanish, German and Italian in NOTEPAD and in READER
`test-utf8`	the code of `VX_UTF8` in a Z80 emulator, against `tools/voscs.py`
`test-view`	`VX_VIEW`: VIEWT (`apps/viewt`) draws through a view of 160 × 96, checked pixel by pixel (`tools/view_check.py`): the view filled, a rectangle cut, 19 of 20 characters, the last pixel, an image in and one out; with 128K: NO EXPANSION
`test-wm`	`VX_WINDOW` and `WM.SYS` (576K): WINA and WINB (`apps/wint`) in windows; seven screenshots checked (`tools/wm_check.py`): one in front of the other, brought to the front by a click with its bar, a click inside from its corner, moved by its title, full screen with F7 and back, closed with X down to the Commander
`test-wm-apps`	CLOCK, SYSINFO and WINA in windows at once (576K); five screenshots checked (`tools/wm_apps_check.py`): the three over each other, CLOCK brought to the front by a click and moved, SYSINFO in front with CLOCK ticking behind it, uncovered, SYSINFO made smaller by its corner
`test-wm-apps2`	DIARY, TELNET and CALC in windows (576K, API 36: the window shows a part of the program's screen); four screenshots checked (`tools/wm_apps2_check.py`): DIARY, TELNET over it, CALC with its buttons in its window, and 1 + 2 = 3 clicked there

`test-wm-notepad`	NOTEPAD in a window with a NOTES.TXT (576K): made smaller by its corner, its columns and rows cut to the window, the status line under them; the page (^M) full screen and the window back where it was (`tools/wm_notepad_check.py`)
`test-wm-pages`	MARKDOWN with INDEX.MD and its preview, READER (wrapped at the window's width) and SHEET (4 columns, 18 rows) in windows (576K); SHEET's page and back (`tools/wm_pages_check.py`)
`test-wm-copies`	windows' copies (576K): WINB moved away from SYSINFO and WINA, over them and away again; WINA and WINB write how many times they have drawn and still say 001, and the Desktop's picture shows under them (`tools/wm_copies_check.py`)
`test-wm-desktop`	the Desktop among the windows (576K, SETUP.CFG with A:'s window): WINA with the Desktop's window and icons behind it; the Desktop (F1) with WINA on its desk; each brought to the front with a click (`tools/wm_desktop_check.py`)
`test-wm-keys`	F5, F6, X and the palette among windows (576K): CLOCK as the accessory, WINA and WINB; the list of programs (F6) and the accessory over the windows, the windows again when they close; WINA's reds (P) leave the colours as they are; WINA's X: ^Q, WINB in front (`tools/wm_keys_check.py`)
`test-mail`	MAIL on a Z80 in Python (`tools/mail_check.py`), with the M4 board of `tools/m4net.py` and POP3 and SMTP servers in Python: the account in `MAIL.CFG`, the newest 20 of 22 messages with their subjects decoded, a plain one, one in quoted-printable UTF-8 and one in parts (base64, not its HTML), scrolling, SAVE AS TEXT, DELETE at QUIT, WRITE and REPLY through SMTP (AUTH LOGIN, "." doubled, the Greek subject in Q), a refusal, a wrong password, no answer, no board
`test-ftp`	FTP on a Z80 in Python (`tools/ftp_check.py`), with an FTP server in Python: anonymous, PASV to the control connection's address, LIST in Unix's and DOS's lines, RETR under an 8.3 name, CWD and CDUP, STOR in pieces, DELE, over 16K not saved, a wrong password, no board
`test-mail-m4`, `test-ftp-m4`	MAIL and FTP in the 1984 emulator with an M4 board (scripts 60 and 61), with the servers `tools/mail_server.py` and `tools/ftp_server.py` on this PC: the list, a message, a `.TXT` sent; FTP's list, a file onto the card and one from it
`test-styles`	STYLES.VBN (`blocks/examples/styles.vbt`) from the Commander: plain, bold, italic, underlined and all three, with the STYLE block
`test-music-block`	with 576K: TUNE.VBN (`blocks/examples/tune.vbt`) starts MUSIC (from B:) behind it with AGANAMEM.MUS; F6 shows "> TUNE" and MUSIC, and RETURN brings MUSIC to the front: A:AGANAMEM, PLAYING. With `MBLOCK_RAM=128`: "CLOSE ANOTHER PROGRAM FIRST"
`test-amsdos`, `test-reboot`, `test-system`	V-OS files in BASIC, reboot, SYSTEM disk

New script:

1. Add an `IF TEST_SCRIPT == n` at the start of the chain in `test_script.asm`, with `ELSE` before the next one and one more `ENDIF` at the end.
2. Add a target to the Makefile, like `test-sdk`.

Each key is written as `dw from, to : db row, bit` of the keyboard. The key number is row·8 + bit, e.g. RETURN = 18 = row 2, bit 2.

Without an emulator: a program's code also runs on a Z80 inside Python (the `z80` package). You load the `.BIN` at `#4000` and make the API calls return immediately (or do whatever is needed, e.g. files in a dictionary). Then you call init and event and read the memory. It is much faster than screenshots for errors in numbers, files and jumps. The label addresses come from `rasm … -s -sl` (a `.sym` file). The network programs' `VX_M4` calls are answered by `tools/m4net.py`, an M4 board in Python with servers in Python at the other end.

Every milestone

1. Code, and a test in Caprice32: new or existing, with screenshots.
2. The tests the change touches.
3. A section in `MILESTONES.md` and `MILESTONES.el.md`, and a "Υλοποίηση Μ..." (implementation) section in `docs/REQUIREMENTS.md`.
4. **Update the manual in all six languages** (`docs/manual/` in Greek, `docs/manual/<code>/` for en, fr, es, de, it, and `sdk/README.md`, `sdk/README.<code>.md` if the API changes), then `make docs` for all the PDFs.
5. Commit and push.

9. On a real CPC

V-OS has only been tested in emulators: Caprice32, 1984 and CPCEMU. A real CPC can behave differently in its disc drive, its keyboard, its CRTC, its monitor, or with an AMX mouse. This chapter is a list of things to try, and what each should do. If something does something else, please tell us. Even "everything worked" helps.

Getting V-OS onto the CPC

The release has two disc images, `vos.dsk` (A:) and `apps.dsk` (B:). Each fits on a 3" disc (180K). Some ways to use them:

- **Gotek or HxC** in place of a drive: copy the `.dsk` files to its USB stick.
- **A PC with a 3.5" drive** connected to the CPC as drive B:: write the images to discs (for example with SAMdisk or DiskImager on Windows).
- **An M4 board:** copy the images to its SD card and mount them from its menu.

Then, in BASIC: `RUN"VOS`.

What to write down first

- The model (CPC 6128, or a 6128 Plus), and if you know it the CRTC type (0 to 4).
- The memory expansion (none, 64K, 512K...), the drives, the mouse, an M4 board.
- The monitor (CTM644 colour, GT65 green, a TV through a converter...).

The list

Each line: what to do, and what you should see.

Starting - [ ] `RUN"VOS` from A:. In a few seconds: the desk, the menu bar at the top, the drive icons on the right. Nothing flickers or jumps. - [ ] The screen fills the whole tube (overscan, 384×272), with nothing cut off at the sides or at the bottom. - [ ] HELP > ABOUT V-OS shows the version of the release. - [ ] SYSINFO (in the Launcher, F2) shows the memory you have.

The disc drive - [ ] TAB on the Desktop opens A:'s window, with its files as icons. A second TAB: B:. - [ ] In the Commander (F3), copy a file from A: to B: (^C). Then leave V-OS (CONTROL+Q) and, in BASIC, `CAT` on B: shows it and `LOAD` reads it. - [ ] Rename (^N) and delete (^D) a file on B:. - [ ] FORMAT DISC (DATA) in the Commander's FILE menu, on a disc you do not need. Then `CAT` in BASIC: an empty disc. - [ ] A disc in SYSTEM format (the CP/M one) is read too. - [ ] Setup (F4): change the language, ^S. Start again with `RUN"VOS`: V-OS starts in that language. - [ ] Without a disc in B:, the Commander says so, and V-OS does not hang.

The keyboard - [ ] Every key in NOTEPAD: letters, digits, symbols, SHIFT, CAPS LOCK, the arrows, DEL, CLR, COPY. - [ ] CONTROL+SHIFT changes the keyboard's language (EN, EΛ, ..., on the right of the menu bar). Greek and accented letters come out right. - [ ] Holding a key repeats it at the speed of Setup's REPEAT DELAY and REPEAT SPEED. - [ ] The function keys: F1 Desktop, F2 Launcher, F3 Commander, F4 Setup, F6 the programs running.

The AMX mouse - [ ] Setup: AMX MOUSE on. The pointer follows the mouse at the same speed across and down, in mode 1 (the Desktop) and in mode 0 (the Launcher). - [ ] If it jumps or goes the wrong way, try the other AMX READING setting. - [ ] Left button: a click; on a window's title, drag it; the corner below on the right, resize it.

Memory - [ ] Without an expansion: NOTEPAD and PONG together, F6 between them; STUDIO runs

on its own. - [ ] With an expansion: four programs at once, each where it was left. PAINT with a big picture. A picture on the desk (Setup, BACKGROUND).

The screen - [ ] PAINT and the Launcher in mode 0 (16 colours): the colours are right. - [ ] READER or NOTEPAD with a page in mode 2 (80 columns): legible, also on a colour monitor. - [ ] Studio's mixed screen (mode 0 above, mode 1 below): the split line stays still, nothing flickers. - [ ] The dark theme (Setup, THEME).

Sound - [ ] MUSIC plays a song; it keeps playing while you go to the Desktop and start another program. - [ ] The games (VROUVES, BRICKS, PONG) have their sounds.

The M4 board (if you have one) - [ ] TAB on the Desktop goes on from B: to DISK, the SD card, with its folders. - [ ] NOTEPAD saves a file on DISK, and opens it again. - [ ] The menu bar shows the M4's time; FILE INFO on a file on DISK gives its date. - [ ] WEBGET downloads a file; READER opens a page from the web. - [ ] TELNET connects to a server (a BBS) and shows what it sends. - [ ] MAIL gets messages from a POP3 server and sends one through SMTP. - [ ] FTP lists a server's folder, gets a file and puts one.

Speed (how long, with a watch) - [ ] From `RUN"VOS` to the desk. - [ ] From F2 to the Launcher, and from a program's icon to the program. - [ ] SHEET: BUDGET.VSH open and worked out.

Telling us

Open an issue at https://github.com/vaspervnp/V-OS-CPC/issues with:

- what you wrote down first (the model, the memory, the monitor...);
- the lines that did something else, and what they did;
- a photo of the screen, if something looks wrong.